

Week 2: Fock Space and symmetric functions

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Our goal today is to study a noncommutative algebra $\mathcal{A}$ (and its extensions) which acts on some vector space $\mathcal{F}$, called Fock space. We will have a decomposition

$$\mathcal{F} = \bigoplus_{\ell \in \mathbb{Z}} \mathcal{F}_\ell \text{ such that each } \mathcal{F}_\ell \cong \Lambda_{\mathbb{C}}.$$

$\mathcal{A}$ is the associative algebra generated by

$$\{\psi_k, \psi_k^* \mid k \in \mathbb{Z} + \frac{1}{2}\}$$

subject to relations

$$\begin{aligned} [\psi_r^*, \psi_s]_+ &= \delta_{rs} \\ [\psi_r, \psi_s]_+ &= [\psi_r^*, \psi_s^*]_+ = 0. \end{aligned}$$

where $[a, b]_+ = ab + ba$.

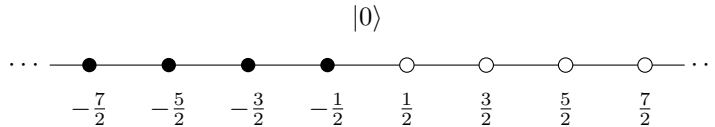
1 Fock space

Definition 1.1. A **state** is an assignment of one of $\{\text{particle}, \text{hole}\}$ to each site in $\mathbb{Z} + \frac{1}{2}$. In other words, a state is

$$\mathbb{Z} + \frac{1}{2} \rightarrow \{\text{particle}, \text{hole}\}.$$

Example 1.2. Denote by $|0\rangle$ the state

$$\begin{cases} k \mapsto \text{particle} & k < 0 \\ k \mapsto \text{hole} & k > 0 \end{cases}$$



Definition 1.3. A state is **good** if it differs from $|0\rangle$ at finitely many sites.

The **Fock space** $\mathcal{F}$ is the complex vector space spanned by the good states.

The action of $\mathcal{A}$ on $\mathcal{F}$ is determined by the actions of the generators ψ and ψ^* on a state.

ψ_k “destroys a particle” at site k , up to sign

ψ_k^* “creates a particle” at site k , up to sign.

More precisely,

$$\psi_k(S) = \begin{cases} 0 & S \text{ has a hole at } k \\ (-1)^{\# \text{ particles to the right of } k} S' & S \text{ has a particle at } k \end{cases},$$

where S' coincides with S except at position k , where it has a hole. And

$$\psi_k^*(S) = \begin{cases} (-1)^{\# \text{ particles to the right of } k} S' & S \text{ has a hole at } k \\ 0 & S \text{ has a particle at } k \end{cases},$$

where S' coincides with S except at position k , where it has a particle. Note that the signs enforce the relation

$$\psi_r^* \psi_s + \psi_s \psi_r^* = \delta_{rs} \text{Id}.$$

1.1 Another description

Now let V be the vector space with basis $\{v_k \mid k \in \mathbb{Z} + \frac{1}{2}\}$. If S has particles at $s_1 > s_2 > s_3 > \dots$, then we can identify S with the vector

$$v_{s_1} \wedge v_{s_2} \wedge \dots \in \bigwedge^{\infty/2} V.$$

Here $\bigwedge^{\infty/2} V$ is the “half-infinite” exterior power, whose elements are sums of ones given by good states.

We then have

$$\begin{aligned} \psi_k^*(S) &= v_k \wedge S \\ \psi_k(S) &= \iota_{v_k}^*(S), \end{aligned}$$

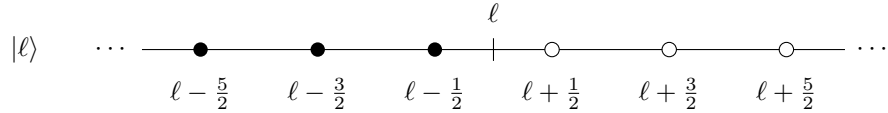
where $\iota_{v_k}^*$ is contraction with the dual basis vector v_k^* .

1.2 Partitions

Observe that a state is good if and only if a state can be obtained, up to sign, from $|0\rangle$ after a finite sequence of ψ_k or ψ_k^* .

In particular, consider the state $|\ell\rangle$ defined by

$$|\ell\rangle = \begin{cases} \psi_{\ell-\frac{1}{2}}^* \psi_{\ell-\frac{3}{2}}^* \dots \psi_{\frac{1}{2}}^* |0\rangle & \ell > 0 \\ \psi_{\ell+\frac{1}{2}} \psi_{\ell+\frac{3}{2}} \dots \psi_{-\frac{1}{2}} |0\rangle & \ell < 0 \\ |0\rangle & \ell = 0 \end{cases}$$



Fix a partition $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$. Define

$$|\lambda; \ell\rangle := \psi_{\ell+\lambda_1-\frac{1}{2}}^* \psi_{\ell+\lambda_2-\frac{3}{2}}^* \dots \psi_{\ell+\lambda_n-n+\frac{1}{2}}^* |\ell-n\rangle.$$

We also define

$$\begin{aligned} |\emptyset; \ell\rangle &:= |\ell\rangle \\ |\lambda\rangle &:= |\lambda; 0\rangle. \end{aligned}$$

We put particles on the vertical edges of the boundary and holes on the horizontal edges; see Figure 1. The corner in column j and row i has label $\ell + j - i$, and the edges have half-integer labels.

We defined $|\lambda; \ell\rangle$ by inserting particles. We could instead define it by inserting holes:

$$|\lambda; \ell\rangle = (-1)^{|\lambda|} \psi_{\ell-\lambda'_1+\frac{1}{2}} \dots \psi_{\ell-\lambda'_m+m-\frac{1}{2}} |\ell+m\rangle$$

where $\lambda' = \lambda^{\text{Tr}}$ and $m = \lambda_1$.

Every good state is equal to $|\lambda; \ell\rangle$ for a unique partition λ and integer ℓ . How do we recover ℓ ?

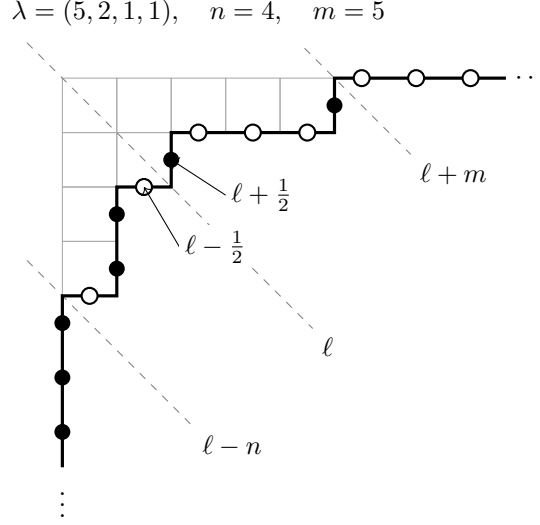


Figure 1: A depiction of $|(5, 2, 1, 1); \ell\rangle$.

Definition 1.4. The **normal ordering** of two operators is

$$:\psi_j^* \psi_k: = -:\psi_k \psi_j^*: = \begin{cases} \psi_j^* \psi_k & j > 0 \\ -\psi_k \psi_j^* & j < 0 \end{cases}.$$

Note that

$$:\psi_j^* \psi_j: = \begin{cases} \mathbf{1}_{\text{particle at } j} & j > 0 \\ -\mathbf{1}_{\text{hole at } j} & j < 0. \end{cases}$$

Fix a good state S . Then $:\psi_j^* \psi_j:(S)$ is nonzero for finitely many j . Hence we may define

$$J_0 := \sum_{j \in \mathbb{Z} + \frac{1}{2}} :\psi_j^* \psi_j:.$$

It turns out that $J_0(S)$ is always an integer multiple of S .

Lemma 1.5.

$$J_0(S) = (\# \text{ particles} > 0 - \# \text{ holes} < 0)S$$

Moreover,

$$J_0 |\lambda; \ell\rangle = \ell |\lambda; \ell\rangle.$$

We have deduced that $\mathcal{F}$ has a basis

$$\{|\lambda; \ell\rangle \mid \lambda \text{ is a partition and } \ell \in \mathbb{Z}\}.$$

Let $\mathcal{F}_\ell$ be the span of $\{|\lambda; \ell\rangle \mid \lambda \text{ is a partition}\}$. Then

$$\mathcal{F} = \bigoplus_{\ell \in \mathbb{Z}} \mathcal{F}_\ell.$$

Each of the subspaces is preserved by the charge-preserving bilinears $\psi_r^* \psi_s$.

1.3 $\mathfrak{gl}(\infty)$

Let $\mathfrak{gl}(\infty)$ be the Lie algebra of matrices with rows and columns indexed by $\mathbb{Z} + \frac{1}{2}$ that is spanned by the identity matrix and matrices with finitely many nonzero entries.

Then there is an action of $\mathfrak{gl}(\infty)$ on $\bigwedge^{\infty/2} V = \mathcal{F}$. It is given by (when A has finitely many nonzero entries)

$$A(v_{s_1} \wedge v_{s_2} \wedge \cdots) = \sum_i v_{s_1} \wedge \cdots \wedge A v_{s_i} \wedge \cdots.$$

Equivalently, the matrix A_{rs} with a 1 in row r and column s acts via $\psi_r^* \psi_s$. The identity matrix acts by the identity. Normal ordering changes this action by a scalar:

$$:\psi_r^* \psi_s: = \psi_r^* \psi_s - \delta_{rs} \mathbf{1}_{r < 0} \text{Id}.$$

The normally ordered operators give a representation of a central extension.

1.4 $\mathcal{F}_\ell$ are irreducible representations

Define

$$J_n := \sum_{r \in \mathbb{Z} + \frac{1}{2}} :\psi_{r-n}^* \psi_r:.$$

The sum is finite on each good state. For $n > 0$, J_n moves some particle n steps left and multiplies by a scalar. When $n < 0$, J_n moves a particle $-n$ steps to the right. Here we sum over all possible moves.

Proposition 1.6. $[J_m, J_n] = m\delta_{m,-n} \text{Id}$.

Each $\mathcal{F}_\ell$ is an irreducible representation of the Heisenberg algebra generated by the J_n and Id .

Define the generating function

$$H[t] = \sum_{q=1}^{\infty} t_q J_q.$$

Then $H[t]$ acts on $\mathcal{F} \otimes_{\mathbb{C}} \mathbb{C}[t_1, t_2, \dots]$. Moreover, the action of $\exp(H[t])$ on $\mathcal{F} \otimes_{\mathbb{C}} \mathbb{C}[t_1, t_2, \dots]$ is well-defined since the J_q commute when q is positive and lower the size of the partition by q .

Theorem 1.7. *There are isomorphisms*

$$\mathcal{F}_\ell \xrightarrow{\sim} \mathbb{C}[t_1, t_2, \dots] \xrightarrow{\sim} \Lambda_{\mathbb{C}}.$$

Here $\Lambda_{\mathbb{C}}$ is the ring of symmetric functions, and $\langle \ell |$ extracts the coefficient of $|\ell\rangle$. The first map is

$$|s\rangle \mapsto \langle \ell | \exp(H[t]) |s\rangle.$$

The second map sends t_i to p_i/i , where p_i is the power sum symmetric function. The composition sends $|\lambda; \ell\rangle$ to the Schur function s_λ .

For $q > 0$,

$$J_q |\lambda\rangle = \sum_{\substack{\lambda/\mu = \text{border strip} \\ \text{of size } q}} (-1)^{\text{ht}(\lambda/\mu)} |\mu\rangle.$$

Here $\text{ht}(\lambda/\mu)$ is the number of rows of the border strip minus one.

AI disclosure

AI was used for editing and created the figures based on the ones drawn during the talk.